# INFT2051 – Final App Documentation

**Details**

| | |
|---|---|
| **Project Name** | HealthLog |
| **Student Name** | WANG ZUZAN |
| **Student Number** | C3552364 |

Please specify the following:

| | | | |
|---|---|---|---|
| **Which Platform did you develop for? (Please Select)** | Android | iOS | Windows |
| **Which emulator should we use to mark? (Please Select)** | Android | iOS | Windows |

**Features Used**

Please Check all that apply:

| | | | |
|---|---|---|---|
| Accelerometer | | Local Storage/Cloud Storage | √ |
| Audio/Video | | GPS/Location | |
| Biometric sensors | | Gyroscope | |
| Bluetooth | | NFC | |
| Camera | | Payment | |
| Compass/Magnetometer | | Security | |
| Existing Libraries (list below) | √ | Other features not listed here (list below) | √ |

| | |
|---|---|
| **Existing Libraries Used:**<br>Microsoft.Data.Sqlite, .NET MAUI framework | **Other Features Used:**<br>**Food nutrition estimation, BMI calculation, personalised nutrition targets, previous record management, background colour setting, font size setting, vibration feedback, screen rotation support, and food input auto-complete/suggestion functionality.** |

**Brief Project Description [100 words maximum]**

*[Here you should write a brief description of the core purpose of your app no more than 100 words].*

HealthLog is an application designed to track nutritional information using the .NET MAUI and C# programming languages. HealthLog enables the user to sign into the application, add their food items and gram weights, and estimate their nutrition, which includes calories, proteins, carbohydrates, fats, and water content. In addition, the home screen contains the feature for automatic completion of food items entered by the user, feedback via pop-ups, vibration alerts, and the date updated based on the system's current date. Users have the ability to edit, view, and delete past records of their food items.

**Project Planning [1000 words maximum]**

*[Here you should explain the:*

- *The features of the app that were able to implement*
- *The native features of the app that you were able to implement*
- *What features/aspects were not included in the final submission*
    - *Note there should be a limited set of missing things, significant gaps in the submission will result in reduced marks. You should also specify the lines of code (or other material e.g., files) that do not work.*

*The word limit here is 1000 words – but this is an upper bound meaning if the assignment works exactly as intended then you can write 'Project works as intended.']*

HealthLog application functions as designed and comprises all key elements included in its plan. In terms of its functionality, the application consists of the following major components: login page, home page, previous records page, and goals/settings page. Home page is a functional core of the application. Users are able to input a product name and its weight in grams, after which estimation of its nutrients is performed: calories, proteins, carbohydrates, fats, and water are calculated. For convenience sake, I included a real-time food suggestion mechanism into the application. This means that as soon as users input the first letters of a product name, the application will offer a variety of suggestions, for example, "banana". This allows reducing number of typos and increasing usability of the application.

After estimating nutrients, the application displays an estimation result on the home page. Moreover, it provides popup feedback that informs users about the fact that estimation is finished. Once the popup is closed, vibration feedback is provided on the majority of Android-based devices.

Furthermore, there is a page for previous records. The users can record their food after estimation and then see the records on the application. This feature enables users to record their previous food instead of seeing one record each time. In addition, users have the capability to delete unneeded records. This app uses SQLite local storage where records will be stored. As a result, even when users close and reopen the app, they will still be able to access the stored data.

The Goals & Settings page will give the user an option to add their personal body data and compute their BMI. Besides this, users can select their nutrition goal depending on what they prefer, either cutting, bulking, or maintaining their weight. Each goal has a different nutrition target. Once users set their chosen goal, the nutritional target values will appear on the homepage. As an illustration, the homepage may display the user's current intake compared to the nutritional target value, for instance, calories out of 1750 Kcal.

IOther personalization options were also added to the app. The user can select the background color and font size as per their choice. This personalizes the user experience according to their preferences. Screen rotation was also supported by the app. This meant that regardless of whether the phone was held vertically or horizontally, the layout could still be accessed. The date mentioned within the app is always the same as the system date.

SQLite database and vibration support are the most significant mobile/native features that have been realized in the application. SQLite is used for storing food record history of users locally. Vibration feedback is provided for some important events when the app performs specific tasks. Support for screen orientation helps to increase adaptability of the application.

There are no particular advanced features incorporated into the app. Such features as cloud accounts sync, barcode scanner, or complete database of foods for analysis are not present in the application. This was done to ensure the focus on the development of the minimum viable product, which includes all essential features for health tracking. In addition, my efforts were aimed at developing an easy-to-use and reliable app; therefore, no complicated functionality has been added to the product. The submitted version provides such basic features as local data storage, food calculation, setting of goals, BMI calculation, customization of the interface, and simple interaction features.

**Audience Justification [500 words maximum]**

*[Here you should justify why your app is appropriate for your audience. Does it meet their needs? If yes how so? Hint: find reviews online to justify or other materials to justify the design of your app.]*

The target audience for my app is students, newbies in fitness, and individuals interested in a straightforward way of recording their food and nutrients daily. This audience was chosen because they might not require an extensively professional health application. Most of them prefer applications that are easy to navigate and understand, where they can add food, view its nutritional value, and determine whether the food fits into their goal.

These requirements were selected after analyzing user needs and other existing applications that help in recording nutrients consumed by users. The App Store description of MyFitnessPal indicates that most nutrition tracking applications allow users to enter foods, calories, macronutrients, water intake, and personalized goals. It helps me understand that recording food names alone is insufficient for users. Users require more than just adding food; they also desire to record the nutritional value of the entered food and compare it with their goals.

The second type of analysis was based on user reviews of Lose It! Users mentioned that the app is simple to use and learnable. Furthermore, they said that Lose It! keeps them committed to achieving their food-related goal. As a result, my intention in designing HealthLog was to create an application that is easy to use for newcomers. I did not want my app to have more than three pages for users to navigate through. Besides, I also intended to incorporate simple buttons like Home, Previous Records, and Goals & Settings. Once users are on the home page, they will have to enter the food name and its weight in grams. The app will then calculate the nutrition content.

Moreover, HealthLog caters to users who have diverse body goals. The users are allowed to enter their body parameters, compute BMI, and select among other options whether they wish to maintain weight, bulk, or cut. After saving the goal, the nutrition targets will be displayed on the homepage of HealthLog where the users can compare their input against the recommended values. Additionally, users can keep past records, check them anytime, and remove those that they no longer require.

In general, I would say that HealthLog fits well into my intended audience as it has all the necessary features like simple food logging, personal nutrition goals, easy navigation, and basic customization. It suits their needs because they need a fast way of checking their nutrition intake, past records, and setting goals in an easy non-professional way.

**Use of AI/GenAI [500 words maximum]**

*[Here you should detail the use of AI and your explanation of the code, here it would also be good to highlight any changes you made along with a short reflection of whether AI generated code that was actually useful]*

*[If you didn't use AI – please outline the reasons why].*

However, the application of AI in this project was limited and involved only specific sections of the code. Specifically, AI assistance was used to write some pieces of code for the function that suggests foods entered by the users, the current date from the system, and improved code formatting and commenting. However, the AI-written code was below the permissible limits and was not applied to the full project. Besides, I personally tested and examined the code before implementing it into the app.

In relation to the function of suggesting foods entered by users, AI assisted in comprehending how I could develop the logic that would be able to compare the user's input with the list of stored foods. For example, when a user enters the first letter of a food item, the app will suggest other options such as "banana". The functionality was modified for my project and related to the estimation process of the home page.

Regarding the second case, AI was useful for comprehending how the date could be shown depending on the system time.

AI was also employed to assist in improving code legibility, including improved comments, names, and formatting. Thus, the AI helped me make my code more legible. However, I did not copy code blindly. Instead, I tested the application in Visual Studio, compared the results to my requirements, and edited the code accordingly.

In general, AI served as an auxiliary instrument that provided only some explanations, formatting, and coding assistance. The overall application architecture, design of pages, testing, and decisions on their implementation were developed and verified independently of the AI system.

**Design Documents [1000 words maximum]**

*[Here you can include any updated diagrams, storyboards, data models etc.]*

HealthLog is designed as a simple mobile application for novice users. I divided it into four main screens: Login, Home, Previous Records, and Goals & Settings. Each screen has its own role. The Login screen allows users to access the app. The Home screen is used for food input and nutrient calculation. The Previous Records screen lets them view and delete saved food records. The Goals & Settings screen allows them to manage their body information, nutrition goals, and visual settings.

After logging in, users move from the Login page to the Home page. From there, they can go to the Previous Records page or the Goals & Settings page. They can also return to the Home page when needed. I designed the navigation in this way so they can clearly understand where they can go from each page.

The Home page is the main function page of the app. On this page, users enter a food name and its weight in grams. The app then calculates nutrition values such as calories, protein, carbs, fats, and water. This page is the center of HealthLog because food tracking is one of its core features. It also shows the results on the same page, so users do not need to open another page.

I also added a food auto-complete suggestion feature. When users type the first letters of a food name, the app can show suggestions. For example, if they type "b," banana may appear. This helps them enter food faster and reduces spelling mistakes. It is useful for beginners because they may not know the exact spelling of every food name.

The Home page also uses the system date automatically. Users do not need to enter the date by themselves. I added this to make calorie recording easier and more accurate. After the calculation is finished, the app shows a pop-up notification. When users close it, compatible Android phones can also give vibration feedback. I included this because it gives users a clear response after an action.

The Previous Records page allows users to review saved food intake records. After they calculate the nutrition content of food, they can save the result. Then they can check it later on this page. I added this page because I want them to see their past intake, not only the current one. They can also delete records they no longer need.

The Goals & Settings page makes the app more personal. On this page, users can enter their height and weight, and the app can calculate their BMI. They can also choose a nutrition goal, such as maintenance, bulking, or cutting. Each goal has its own target nutrition values. After they save the goal, the target values appear on the Home page, such as 0/1750 Kcal.

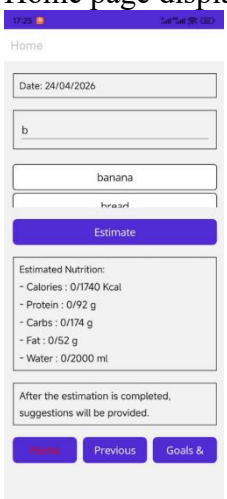
This page also includes visual customization. Users can change the background color and font size. I added these options because some users may want the app to look different. Changing the font size can make it easier to read, while changing the background color can make it feel more personal.

The main data model I used in HealthLog is the food record model. It includes Id, date, record name, calories, protein, carbs, fat, and water. The Id is used as a unique number for each record. The date stores when the record was created. The record name stores the food name. The nutrition values store the estimated result. This model is connected to SQLite local storage. When users save, view, or delete records, the app handles the data locally.

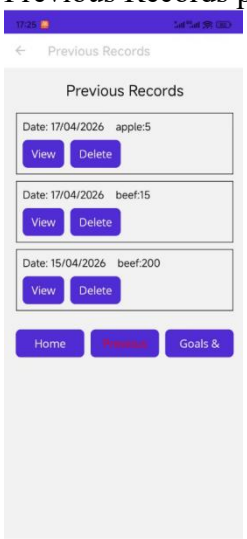
The basic storyboard of the app starts from the Login page. After users enter their login information, they go to the Home page. They enter the food name and weight, and the app may give suggestions while they type. Then they start the estimation process and see the calculated nutrition values. They can save the result if needed. Later, they can go to Previous Records to review or delete old records. If they want to change personal goals, they can go to Goals & Settings, enter their body information, calculate BMI, choose a goal, and save it. After that, the updated nutrition targets will appear on the Home page.

Overall, the design documents show that HealthLog was developed with a clear structure. Its pages, navigation, data model, and user flow work together to support the main purpose of the app.

Home page displaying estimates of nutrition and targets.Food search suggestions.




Previous Records page showing saved food records.




Goals & Setting page showing nutrition target values and save button.

**Future Plans [500 words maximum]**

*[Here you should detail any future plans, stretch goals, or things you intend to do to your app post the final submission. Especially useful for those you intend to post to the App Stores. Here you should detail new features to deploy, how you will monetise your app etc.]*

*[If you do not intend to post to the App Stores give a brief description of the reasons why you are not posting the app to the store].*

Post-delivery of this project, I will proceed to further improve HealthLog. At present, the application has all the required features, including the entry of food information, nutritional content analysis, past data management, BMI measurement, target-setting, user interface customizability, vibration alerting, and local data storage. But in my opinion, the utility of this application will be increased by continued development beyond this academic term.

The first thing that I intend to improve in the future is to increase the database. Right now, the information provided is based mostly on the food entries in the project. My goal for the future is to integrate my application with a bigger and better food or nutrition database, which will help me collect information about a greater variety of foods.

Another improvement that I would like to achieve relates to record analysis. At present, users have the ability to save, view, and delete their past food records from the application. I would like to make improvements such that users could see summary statistics for the week and month, including some visual representations of calories, protein, carbohydrates, and fats consumed.

Another future idea is incorporating some personal health elements. For example, users will be able to make their goals related to target weight, water consumption or any other personal calorie limit. I might also integrate an alarm function that will help users to recall about making records of food consumption or drinking water.

At the same time, it seems rather improbable that I will submit the existing application for publishing at the moment of its submission to the App Store or Google Play. On the contrary, I will test it further, make changes regarding UI and enlarge my data base of food items. In case I decide to release another version in the future, I will consider the option of free basic features and some paid options.

**Reflections on building your app [500 words maximum]**

*[Here you should detail your reflection on the process of building your app, give insight into your successes and challenges, is there anything you would do differently? What worked well?]*

The process of building HealthLog is definitely meaningful to me. At first, before working on this project, I had a misconception that the development of an application was merely some

page designs with the addition of buttons. However, after I started working seriously on it, I realized that this is not just a page design but the process of creating complex connections among pages, designing the storage of data, considering users' ways of interaction with the app and implementing stable functions in general.

As for the strengths that I have shown during the creation process, I think the main strength is the successful implementation of the main functions that I set myself initially. In my app, users are able to log in and log out, write down the names of foods, number of grams, calculate nutrition, save old records, delete them, count BMI, pick their own goals, as well as change background colour and font size. Moreover, I was able to implement additional functions in the form of food inputs, system date, vibrating function and local storage based on SQLite database.

On the other hand, there were some challenges that I faced. The first and foremost challenge was related to creating the connection between different functions. For instance, when a user saves their goal in the Goals and Settings section, their nutrition goal should also change on the Home page. The connection between such pages often gave me trouble. The second problem was related to saving and retrieving data from the app. Some of the records got stored in the database, while displaying incorrectly in another page.

As a result, this project helped me learn both some coding skills and also something more about app development. In particular, I understood that the smallest details, including the button names and popup information, can greatly influence the performance of an application. Now, I am more experienced in using such programming languages as C# and XAML and such tools as MAUI and SQLite.

If I had a chance to redo this project, I would have made plans regarding the structure of the data and the flow of pages earlier on. This would have saved me time in the process. In all, this project helped me grow technically and in terms of thinking of an effective application design.

**References (as applicable)**

*[Here you can add references to any code, images, or other materials that you did not create yourself]*

MyFitnessPal. (n.d.). *MyFitnessPal: Calorie Counter*. Apple App Store. Retrieved from https://apps.apple.com/us/app/myfitnesspal-calorie-counter/id341232718

Lose It! (n.d.). *Lose It! – Calorie Counter ratings and reviews*. Apple App Store. Retrieved from https://apps.apple.com/us/app/lose-it-calorie-counter/id297368629?see-all=reviews

(My github:https://github.com/webyog/sqlyog-community.git)